CHAPTER 3

Organic Compounds: Alkanes and Their Stereochemistry



FIGURE 3.1 The bristlecone pine is the oldest living organism on Earth. The waxy coating on its needles contains a mixture of organic compounds called alkanes, the subject of this chapter. (credit: modification of work "Gnarly Bristlecone Pine" by Rick Goldwaser/Flickr, CC BY 2.0)

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WHY THIS CHAPTER? The group of organic compounds called *alkanes* are simple and relatively unreactive, but they nevertheless provide a useful vehicle for introducing some important general ideas. In this chapter, we'll use alkanes to introduce the basic approach to naming organic compounds and to take an initial look at some of the three-dimensional aspects of molecules, a topic of particular importance in understanding biological organic chemistry.

According to *Chemical Abstracts,* the publication that abstracts and indexes the chemical literature, there are more than 195 million known organic compounds. Each of these compounds has its own physical properties, such as melting point and boiling point, and each has its own chemical reactivity.

Chemists have learned through years of experience that organic compounds can be classified into families according to their structural features and that the members of a given family have similar chemical behavior. Instead of 195 million compounds with random reactivity, there are a few dozen families of organic compounds whose chemistry is reasonably predictable. We'll study the chemistry of specific families throughout much of this book, beginning in this chapter with a look at the simplest family, the *alkanes.*

3.1 Functional Groups

The structural features that make it possible to classify compounds into families are called *functional groups.* A **functional group** is a group of atoms within a molecule that has a characteristic chemical behavior. Chemically, a given functional group behaves in nearly the same way in every molecule it's a part of. For example, compare ethylene, a plant hormone that causes fruit to ripen, with menthene, a much more complicated molecule found in peppermint oil. Both substances contain a carbon–carbon double-bond functional group, and both therefore react with Br_2 in the same way to give a product in which a Br atom has added to each of the double-bond carbons (FIGURE 3.2). This example is typical: *the chemistry of every organic molecule, regardless of size and complexity, is determined by the functional groups it contains.*

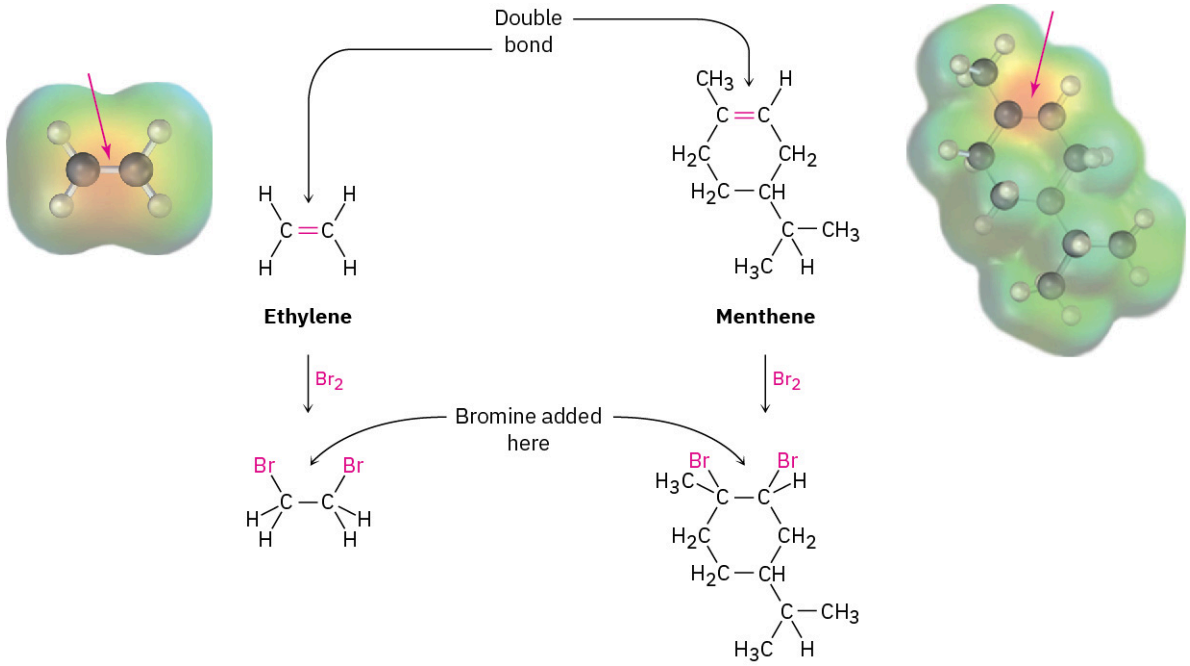


FIGURE 3.2 **The reactions of ethylene and menthene with** bromine. In both molecules, the carbon–carbon double-bond functional group has a similar polarity pattern, so both molecules react with Br_2 in the same way. The size and complexity of the molecules are not important.

Look at TABLE 3.1, which lists many of the common functional groups and gives simple examples of their occurrence. Some functional groups have only carbon–carbon double or triple bonds; others have halogen atoms; and still others contain oxygen, nitrogen, or sulfur. Much of the chemistry you'll be studying is the chemistry of these functional groups.

Functional Groups with Carbon–Carbon Multiple Bonds

Alkenes, alkynes, and arenes (aromatic compounds) all contain carbon–carbon multiple bonds. *Alkenes* have a double bond, *alkynes* have a triple bond, and *arenes* have alternating double and single bonds in a six-membered ring of carbon atoms. They look different, but because of their structural similarities, they also have chemical similarities.

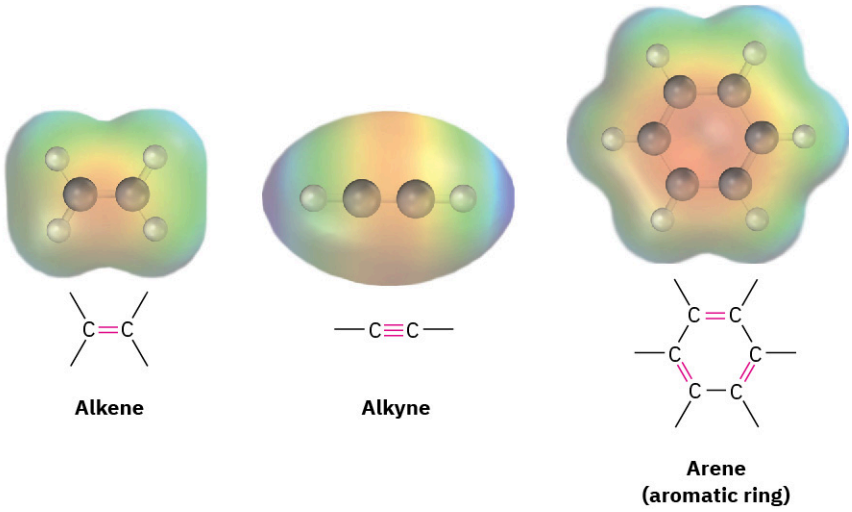


TABLE 3.1 Structures of Some Common Functional Groups

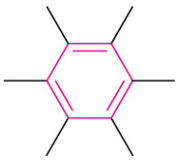
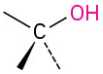
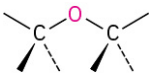
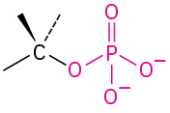
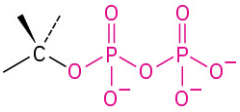
Name	Structure*	Name ending	Example
Alkene (double bond)		*-ene*	$H_2C{=}CH_2$ Ethene
Alkyne (triple bond)	–C≡C–	*-yne*	HC≡CH Ethyne
Arene (aromatic ring)		None	 Benzene
Halide	 (X=F, Cl, Br, I)	None	CH_3Cl Chloromethane
Alcohol		*-ol*	CH_3OH Methanol
Ether		*ether*	CH_3OCH_3 Dimethyl ether
Monophosphate		*phosphate*	$CH_3OPO_3^{2-}$ Methyl phosphate
Diphosphate		*diphosphate*	$CH_3OP_2O_6^{3-}$ Methyl diphosphate

TABLE 3.1 Structures of Some Common Functional Groups

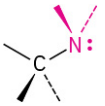
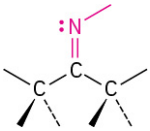
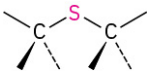
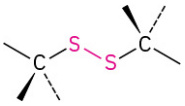
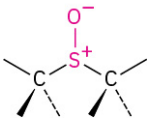
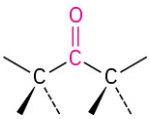
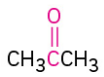
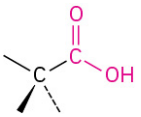
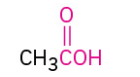
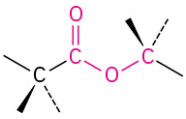
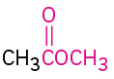
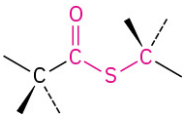
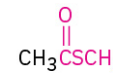
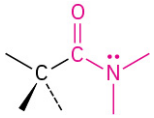
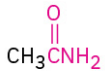
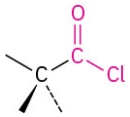
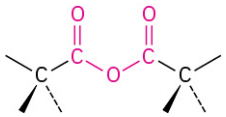
Name	Structure*	Name ending	Example
Amine		*-amine*	CH_3NH_2 Methylamine
Imine (Schiff base)		None	 Acetone imine
Nitrile	–C≡N	*-nitrile*	$CH_3C{\equiv}N$ Ethanenitrile
Thiol		*-thiol*	CH_3SH Methanethiol
Sulfide		*sulfide*	CH_3SCH_3 Dimethyl sulfide
Disulfide		*disulfide*	CH_3SSCH_3 Dimethyl disulfide
Sulfoxide		*sulfoxide*	 Dimethyl sulfoxide
Aldehyde		*-al*	 Ethanal
Ketone		*-one*	 Propanone
Carboxylic acid		*-oic acid*	 Ethanoic acid
Ester		*-oate*	 Methyl ethanoate
Thioester		*-thioate*	 Methyl ethanethioate

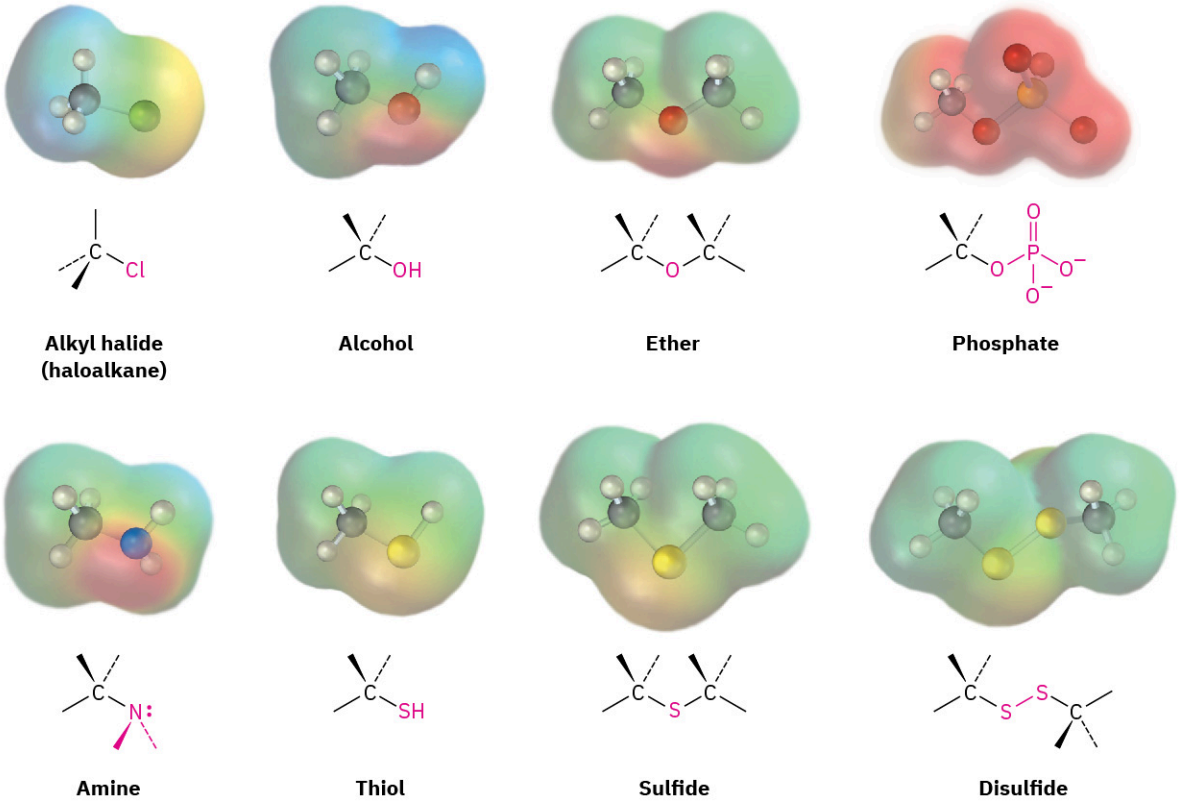
TABLE 3.1 Structures of Some Common Functional Groups

Name	Structure*	Name ending	Example
Amide		*-amide*	 Ethanamide
Acid chloride		*-oyl chloride*	 Ethanoyl chloride
Carboxylic acid anhydride		*-oic anhydride*	 Ethanoic anhydride

*The bonds whose connections aren't specified are assumed to be attached to carbon or hydrogen atoms in the rest of the molecule.

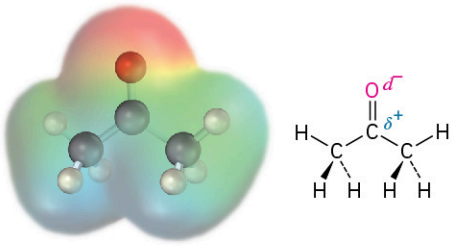
Functional Groups with Carbon Singly Bonded to an Electronegative Atom

Alkyl halides (haloalkanes), alcohols, ethers, alkyl phosphates, amines, thiols, sulfides, and disulfides all have a carbon atom singly bonded to an electronegative atom—halogen, oxygen, nitrogen, or sulfur. **Alkyl halides** have a carbon atom bonded to halogen (–X), **alcohols** have a carbon atom bonded to the oxygen of a hydroxyl group (–OH), **ethers** have two carbon atoms bonded to the same oxygen, organophosphates have a carbon atom bonded to the oxygen of a phosphate group ($-OPO_3^{2-}$), **amines** have a carbon atom bonded to a nitrogen, **thiols** have a carbon atom bonded to the sulfur of an –SH group, **sulfides** have two carbon atoms bonded to the same sulfur, and **disulfides** have carbon atoms bonded to two sulfurs that are joined together. In all cases, the bonds are polar, with the carbon atom bearing a partial positive charge ($\delta+$) and the electronegative atom bearing a partial negative charge ($\delta-$).

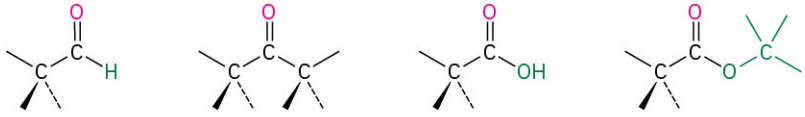


Functional Groups with a Carbon–Oxygen Double Bond (Carbonyl Groups)

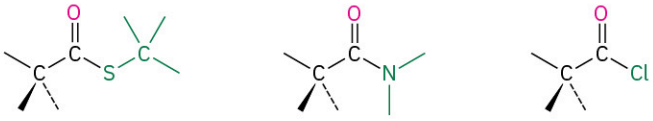
The **carbonyl group**, C=O (pronounced car-bo-**neel**) is common to many of the families listed in TABLE 3.1. Carbonyl groups are present in a majority of organic compounds and in practically all biological molecules. These compounds therefore behave similarly in many respects but differ depending on the identity of the other atoms bonded to the carbonyl-group carbon. **Aldehydes** have at least one hydrogen bonded to the C=O, **ketones** have two carbons bonded to the C=O, **carboxylic acids** have an –OH group bonded to the C=O, **esters** have an ether-like oxygen bonded to the C=O, **thioesters** have a sulfide-like sulfur bonded to the C=O, **amides** have an amine-like nitrogen bonded to the C=O, **acid chlorides** have a chlorine bonded to the C=O, and so on. In all these functional groups, the carbonyl carbon atom bears a partial positive charge (δ+), and the oxygen bears a partial negative charge (δ–).



Acetone—a typical carbonyl compound



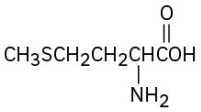
Aldehyde Ketone Carboxylic acid Ester



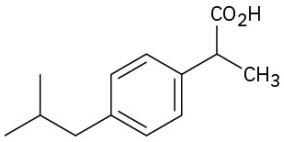
Thioester Amide Acid chloride

PROBLEM 3-1 Use Table 3.1 to identify the functional groups in each of the following molecules:

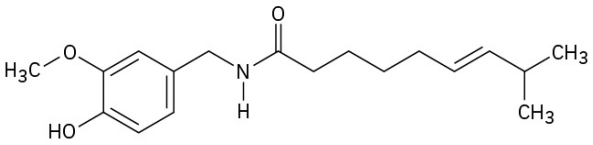
(a) Methionine, an amino acid:



(b) Ibuprofen, a pain reliever:

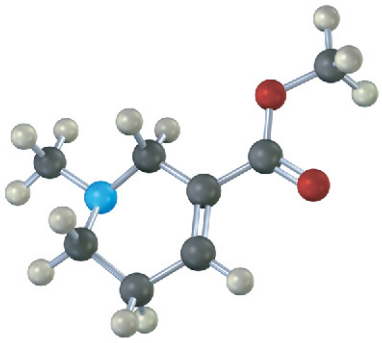


(c) Capsaicin, the pungent substance in chili peppers:



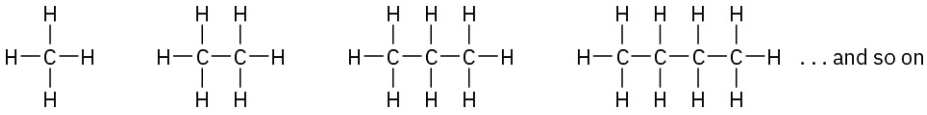
PROBLEM 3-2 Propose structures for simple molecules that contain the following functional groups:
(a) Alcohol **(b)** Aromatic ring **(c)** Carboxylic acid **(d)** Amine **(e)** Both ketone and amine **(f)** Two double bonds

PROBLEM 3-3 Identify the functional groups in the following model of arecoline, a veterinary drug used to control worms in animals. Convert the drawing into a line-bond structure and a molecular formula (red = O, blue = N, black = C, gray = H).



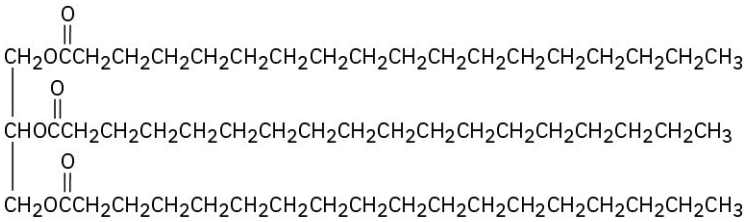
3.2 Alkanes and Alkane Isomers

Before beginning a systematic study of the different functional groups, let's look first at the simplest family of molecules to develop some general ideas that apply to all families. We saw in **Section 1.7** that the carbon–carbon single bond in ethane results from σ (head-on) overlap of carbon sp^3 hybrid orbitals. If we imagine joining three, four, five, or even more carbon atoms by C–C single bonds, we can generate the large family of molecules called **alkanes**.



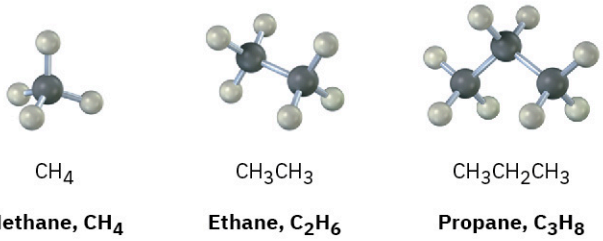
Methane **Ethane** **Propane** **Butane**

Alkanes are often described as *saturated hydrocarbons:* **hydrocarbons** because they contain only carbon and hydrogen; **saturated** because they have only C–C and C–H single bonds and thus contain the maximum possible number of hydrogens per carbon. They have the general formula C_nH_{2n+2}, where n is an integer. Alkanes are also occasionally called **aliphatic** compounds, a name derived from the Greek *aleiphas,* meaning "fat." We'll see in **Section 27.1** that many animal fats contain long carbon chains similar to alkanes.



A typical animal fat

Think about the ways that carbon and hydrogen might combine to make alkanes. With one carbon and four hydrogens, only one structure is possible: methane, CH_4. Similarly, there is only one combination of two carbons with six hydrogens (ethane, CH_3CH_3) and only one combination of three carbons with eight hydrogens (propane, $CH_3CH_2CH_3$). When larger numbers of carbons and hydrogens combine, however, more than one structure is possible. For example, there are two substances with the formula C_4H_{10}: the four carbons can all be in a row (butane), or they can branch (isobutane). Similarly, there are three C_5H_{12} molecules, and so on for larger alkanes.



CH_4

Methane, CH_4

CH_3CH_3

Ethane, C_2H_6

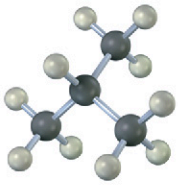
$CH_3CH_2CH_3$

Propane, C_3H_8



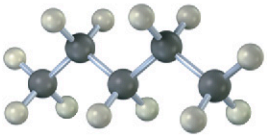
$CH_3CH_2CH_2CH_3$

Butane, C_4H_{10}



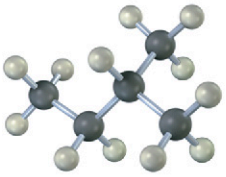
$CH_3CH(CH_3)CH_3$

Isobutane, C_4H_{10} (2-methylpropane)



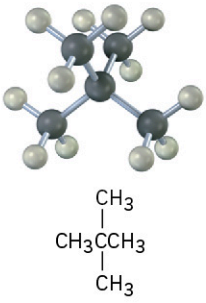
$CH_3CH_2CH_2CH_2CH_3$

Pentane, C_5H_{12}



$CH_3CH_2CH(CH_3)CH_3$

2-Methylbutane, C_5H_{12}



$CH_3C(CH_3)_2CH_3$

2,2-Dimethylpropane, C_5H_{12}

Compounds like butane and pentane, whose carbons are all connected in a row, are called **straight-chain alkanes**, or *normal alkanes.* Compounds like 2-methylpropane (isobutane), 2-methylbutane, and 2,2-dimethylpropane, whose carbon chains branch, are called **branched-chain alkanes**.

Compounds like the two C_4H_{10} molecules and the three C_5H_{12} molecules, which have the same formula but different structures, are called **Isomers**, from the Greek *isos* + *meros,* meaning "made of the same parts." Isomers have the same numbers and kinds of atoms but differ in the way the atoms are arranged. Compounds like butane and isobutane, whose atoms are connected differently, are called **constitutional isomers**. We'll see shortly that other kinds of isomers are also possible, even among compounds whose atoms are connected in the same order. As TABLE 3.2 shows, the number of possible alkane isomers increases dramatically with the number of carbon atoms.

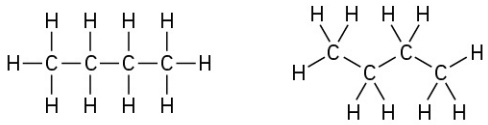
TABLE 3.2 Number of Alkane Isomers

Formula	Number of isomers	Formula	Number of isomers
C_6H_{14}	5	$C_{10}H_{22}$	75
C_7H_{16}	9	$C_{15}H_{32}$	4347
C_8H_{18}	18	$C_{20}H_{42}$	366,319
C_9H_{20}	35	$C_{30}H_{62}$	4,111,846,763

Constitutional isomerism is not limited to alkanes—it occurs widely throughout organic chemistry. Constitutional isomers may have different carbon skeletons (as in isobutane and butane), different functional groups (as in ethanol and dimethyl ether), or different locations of a functional group along the chain (as in isopropylamine and propylamine). Regardless of the reason for the isomerism, constitutional isomers are always different compounds with different properties but with the same formula.

Different carbon skeletons C_4H_{10}	CH_3CHCH_3 (with CH_3 branch)	and	$CH_3CH_2CH_2CH_3$
	2-Methylpropane (isobutane)		**Butane**
Different functional groups C_2H_6O	CH_3CH_2OH	and	CH_3OCH_3
	Ethanol		**Dimethyl ether**
Different position of functional groups C_3H_9N	CH_3CHCH_3 (NH_2)	and	$CH_3CH_2CH_2NH_2$
	Isopropylamine		**Propylamine**

A given alkane can be drawn in many ways. For example, the straight-chain, four-carbon alkane called butane can be represented by any of the structures shown in **FIGURE 3.3**. These structures don't imply any particular three-dimensional geometry for butane; they indicate only the connections among atoms. In practice, as noted in **Section 1.12**, chemists rarely draw all the bonds in a molecule and usually refer to butane by the condensed structure, $CH_3CH_2CH_2CH_3$ or $CH_3(CH_2)_2CH_3$. Still more simply, butane can be represented as *n*-C_4H_{10}, where *n* denotes *normal* (straight-chain) butane.



$CH_3—CH_2—CH_2—CH_3$ $CH_3CH_2CH_2CH_3$ $CH_3(CH_2)_2CH_3$

FIGURE 3.3 Some representations of butane, C_4H_{10}. The molecule is the same regardless of how it's drawn. These structures imply only that butane has a continuous chain of four carbon atoms; they do not imply any specific geometry.

Straight-chain alkanes are named according to the number of carbon atoms they contain, as shown in **TABLE 3.3**. With the exception of the first four compounds—methane, ethane, propane, and butane—whose names have historical roots, the alkanes are named based on Greek numbers. The suffix *-ane* is added to the end of each name to indicate that the molecule identified is an alkane. Thus, *pent*ane is the five-carbon alkane, *hex*ane is the six-carbon alkane, and so on. We'll soon see that these alkane names form the basis for naming all other organic compounds, so at least the first ten should be memorized.

TABLE 3.3 Names of Straight-Chain Alkanes

Number of carbons (*n*)	Name	Formula (C_nH_{2n+2})	Number of carbons (*n*)	Name	Formula (C_nH_{2n+2})
1	Methane	CH_4	9	Nonane	C_9H_{20}
2	Ethane	C_2H_6	10	Decane	$C_{10}H_{22}$
3	Propane	C_3H_8	11	Undecane	$C_{11}H_{24}$
4	Butane	C_4H_{10}	12	Dodecane	$C_{12}H_{26}$
5	Pentane	C_5H_{12}	13	Tridecane	$C_{13}H_{28}$

TABLE 3.3 Names of Straight-Chain Alkanes

Number of carbons (n)	Name	Formula (C_nH_{2n+2})	Number of carbons (n)	Name	Formula (C_nH_{2n+2})
6	Hexane	C_6H_{14}	20	Icosane	$C_{20}H_{42}$
7	Heptane	C_7H_{16}	30	Triacontane	$C_{30}H_{62}$
8	Octane	C_8H_{18}			

WORKED EXAMPLE 3.1

Drawing the Structures of Isomers

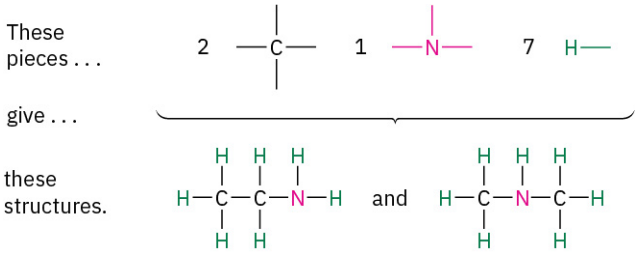
Propose structures for two isomers with the formula C_2H_7N.

Strategy

We know that carbon forms four bonds, nitrogen forms three, and hydrogen forms one. Write down the carbon atoms first, and then use trial and error plus intuition to put the pieces together.

Solution

There are two isomeric structures. One has the connection C–C–N, and the other has the connection C–N–C.



PROBLEM 3-4 Draw structures of the five isomers of C_6H_{14}.

PROBLEM 3-5 Propose structures that meet the following descriptions:

(a) Two isomeric esters with the formula $C_5H_{10}O_2$:

(b) Two isomeric nitriles with the formula C_4H_7N

(c) Two isomeric disulfides with the formula $C_4H_{10}S_2$

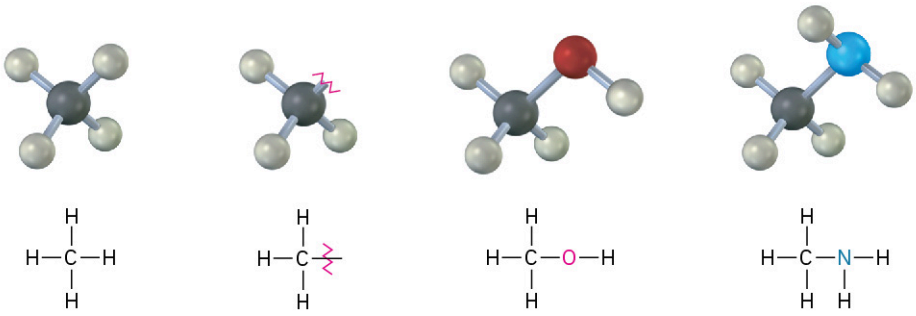
PROBLEM 3-6 How many isomers are there with the following descriptions?

(a) Alcohols with the formula C_3H_8O **(b)** Bromoalkanes with the formula C_4H_9Br

(c) Thioesters with the formula C_4H_8OS

3.3 Alkyl Groups

If you imagine removing a hydrogen atom from an alkane, the partial structure that remains is called an **alkyl group**. Alkyl groups are not stable compounds themselves, they are simply parts of larger compounds and are named by replacing the *-ane* ending of the parent alkane with an *-yl* ending. For example, removal of a hydrogen from methane, CH_4, generates a methyl group, $–CH_3$, and removal of a hydrogen from ethane, CH_3CH_3, generates an ethyl group, $–CH_2CH_3$. Similarly, removal of a hydrogen atom from the end carbon of any straight-chain alkane gives the series of straight-chain alkyl groups shown in TABLE 3.4. Combining an alkyl group with any of the functional groups listed earlier makes it possible to generate and name many thousands of compounds. For example:



Methane **A methyl group** **Methyl alcohol (methanol)** **Methylamine**

TABLE 3.4 Some Straight-Chain Alkyl Groups

Alkane	Name	Alkyl group	Name (abbreviation)
CH_4	Methane	$-CH_3$	Methyl (Me)
CH_3CH_3	Ethane	$-CH_2CH_3$	Ethyl (Et)
$CH_3CH_2CH_3$	Propane	$-CH_2CH_2CH_3$	Propyl (Pr)
$CH_3CH_2CH_2CH_3$	Butane	$-CH_2CH_2CH_2CH_3$	Butyl (Bu)
$CH_3CH_2CH_2CH_2CH_3$	Pentane	$-CH_2CH_2CH_2CH_2CH_3$	Pentyl, or amyl

Just as straight-chain alkyl groups are generated by removing a hydrogen from an end carbon, branched alkyl groups are generated by removing a hydrogen atom from an internal carbon. Two 3-carbon alkyl groups and four 4-carbon alkyl groups are possible (FIGURE 3.4).

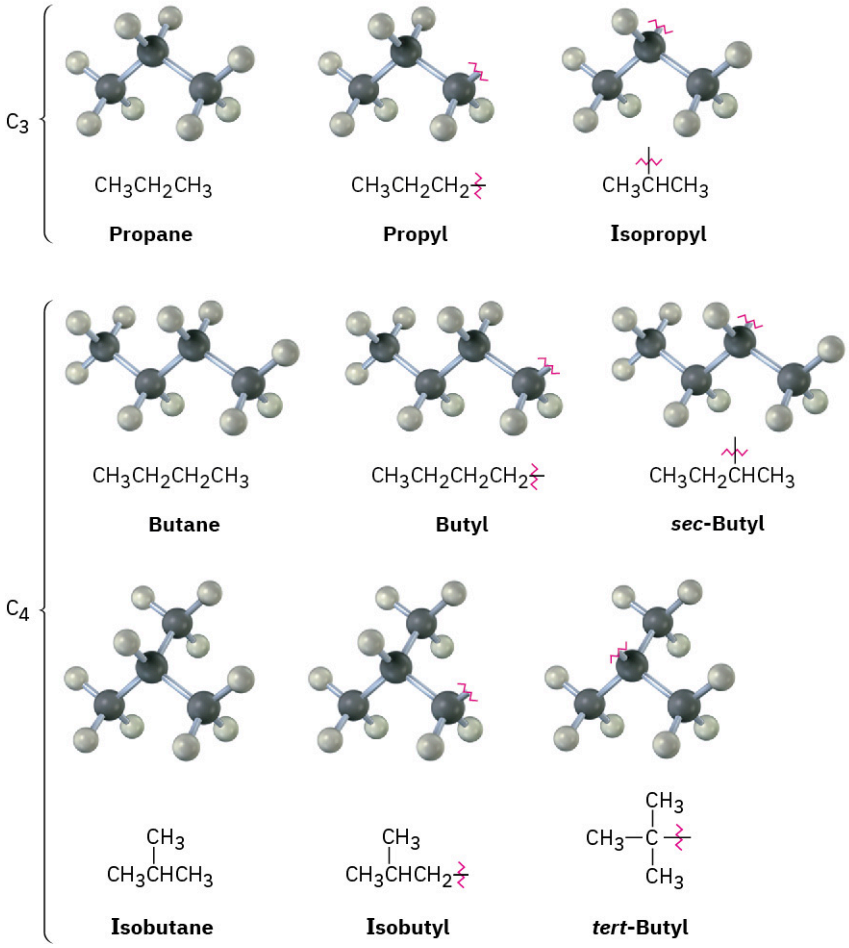


FIGURE 3.4 Alkyl groups generated from straight-chain alkanes.

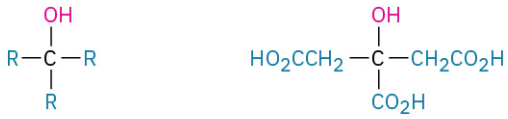
One further comment about naming alkyl groups: the prefixes *sec-* (for secondary) and *tert-* (for tertiary) used for the C_4 alkyl groups in **FIGURE 3.4** refer to the number of other carbon atoms attached to the branching carbon atom. There are four possibilities: primary (1°), secondary (2°), tertiary (3°), and quaternary (4°).



***Primary* carbon (1°) is bonded to one other carbon.** | ***Secondary* carbon (2°) is bonded to two other carbons.** | ***Tertiary* carbon (3°) is bonded to three other carbons.** | ***Quaternary* carbon (4°) is bonded to four other carbons.**

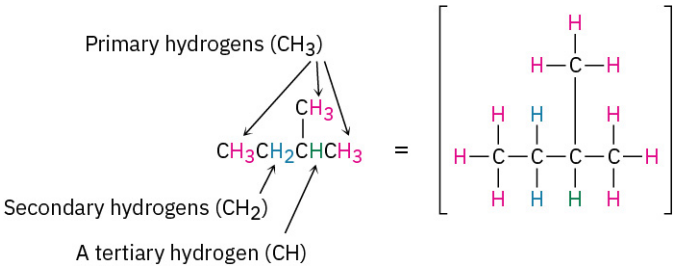
The symbol **R** is used here and throughout organic chemistry to represent a generalized organic group. The R group can be methyl, ethyl, propyl, or any of a multitude of others. You might think of **R** as representing the **R**est of the molecule, which isn't specified.

The terms *primary, secondary, tertiary,* and *quaternary* are routinely used in organic chemistry, and their meanings need to become second nature. For example, if we were to say, "Citric acid is a tertiary alcohol," we would mean that it has an alcohol functional group (–OH) bonded to a carbon atom that is itself bonded to three other carbons.



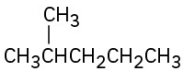
General class of tertiary alcohols, R_3COH | **Citric acid—a specific tertiary alcohol**

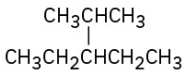
In addition to speaking of carbon atoms as being primary, secondary, or tertiary, we speak of hydrogens in the same way. Primary hydrogen atoms are attached to primary carbons (RCH_3), secondary hydrogens are attached to secondary carbons (R_2CH_2), and tertiary hydrogens are attached to tertiary carbons (R_3CH). There is, however, no such thing as a quaternary hydrogen. (Why not?)

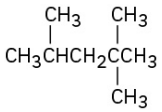


PROBLEM 3-7 Draw the eight 5-carbon alkyl groups (pentyl isomers).

PROBLEM 3-8 Identify the carbon atoms in the following molecules as primary, secondary, tertiary, or quaternary:

(a) 

(b) 

(c) 

PROBLEM 3-9 Identify the hydrogen atoms on the compounds shown in Problem 3-8 as primary, secondary, or tertiary.

PROBLEM 3-10 Draw structures of alkanes that meet the following descriptions:

(a) An alkane with two tertiary carbons **(b)** An alkane that contains an isopropyl group

(c) An alkane that has one quaternary and one secondary carbon

3.4 Naming Alkanes

In earlier times, when relatively few pure organic chemicals were known, new compounds were named at the whim of their discoverer. Thus, urea (CH_4N_2O) is a crystalline substance isolated from urine; morphine ($C_{17}H_{19}NO_3$) is an analgesic (painkiller) named after Morpheus, the Greek god of dreams; and acetic acid, the primary organic constituent of vinegar, is named from the Latin word for vinegar, *acetum*.

As the science of organic chemistry slowly grew in the 19th century, so too did the number of known compounds and the need for a systematic method of naming them. The system of naming (nomenclature) we'll use in this book is that devised by the International Union of Pure and Applied Chemistry (IUPAC, usually spoken as **eye**-you-pac).

A chemical name typically has four parts in the IUPAC system: parent, prefix, locant, and suffix. The **parent** name identifies the main part of the molecule and tells how many carbon atoms are in that part. **Prefixes** identify the various **substituent** groups attached to the parent. **Locants** give the positions of the attached substituents. And the **suffix** identifies the primary functional group attached to the parent.



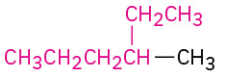
As we cover new functional groups in later chapters, the applicable IUPAC rules of nomenclature will be given. In addition, Appendix A at the back of this book gives an overall view of organic nomenclature and shows how compounds that contain more than one functional group are named. (If preferred, you can study that appendix now.) For the present, let's see how to name branched-chain alkanes and learn some general rules that are applicable to all compounds.

All but the most complex branched-chain alkanes can be named by following four steps. For a very few compounds, a fifth step is needed.

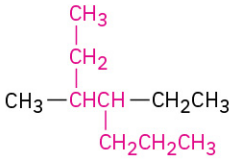
STEP 1

Identify the parent hydrocarbon.

(a) Find the longest continuous chain of carbon atoms in the molecule, and use the name of that chain as the parent name. The longest chain may not always be apparent from the manner of writing; you may have to "turn corners."

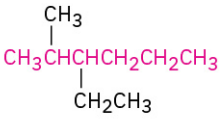


Named as a substituted hexane



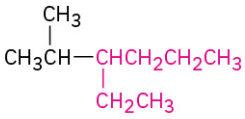
Named as a substituted heptane

(b) If two different chains of equal length are present, choose the one with the larger number of branch points as the parent.



Named as a hexane with *two* substituents

NOT

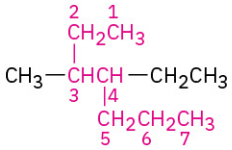


as a hexane with *one* substituent

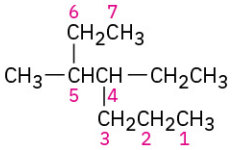
STEP 2

Number the atoms in the longest chain.

(a) Beginning at the end nearer the first branch point, number each carbon atom in the parent chain.

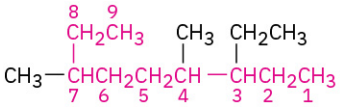


NOT

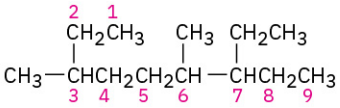


The first branch occurs at C3 in the proper system of numbering, not at C4.

(b) If there is branching an equal distance away from both ends of the parent chain, begin numbering at the end nearer the second branch point.



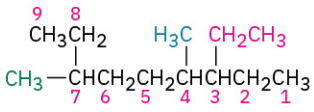
NOT



STEP 3

Identify and number the substituents.

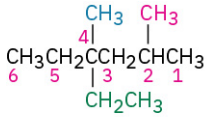
(a) Assign a number to each substituent to locate its point of attachment to the parent chain.



Named as a nonane

Substituents:	On C3, CH_2CH_3	(3-ethyl)
	On C4, CH_3	(4-methyl)
	On C7, CH_3	(7-methyl)

(b) If there are two substituents on the same carbon, give both the same number. There must be as many numbers in the name as there are substituents.



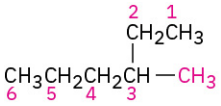
Named as a hexane

Substituents: On C2, CH_3 (2-methyl)
On C4, CH_3 (4-methyl)
On C4, CH_2CH_3 (4-ethyl)

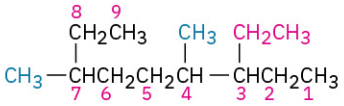
STEP 4

Write the name as a single word.

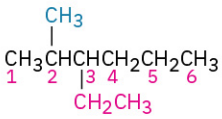
Use hyphens to separate the different prefixes, and use commas to separate numbers. If two or more different substituents are present, cite them in alphabetical order. If two or more identical substituents are present on the parent chain, use one of the multiplier prefixes *di-*, *tri-*, *tetra-*, and so forth, but don't use these prefixes for alphabetizing. Full names for some of the examples we have been using are as follows:



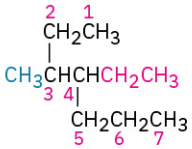
3-Methylhexane



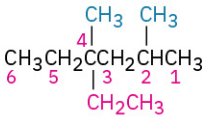
3-Ethyl-4,7-dimethylnonane



3-Ethyl-2-methylhexane



4-Ethyl-3-methylheptane

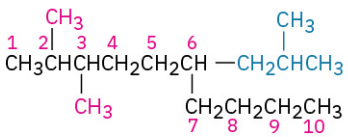


4-Ethyl-2,4-dimethylhexane

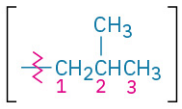
STEP 5

Name a branched substituent as though it were itself a compound.

In some particularly complex cases, a fifth step is necessary. It occasionally happens that a substituent on the main chain is itself branched. In the following case, for instance, the substituent at C6 is a three-carbon chain with a methyl group. To name the compound fully, the branched substituent must first be named.

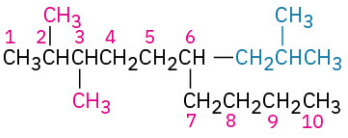


Named as a 2,3,6-trisubstituted decane



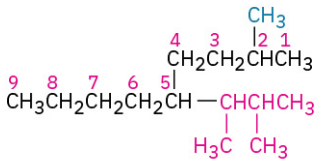
A 2-methylpropyl substituent

Number the branched substituent beginning at the point of its attachment to the main chain, and identify it—in this case, a 2-methylpropyl group. The substituent is treated as a whole and is alphabetized according to the first letter of its complete name, including any numerical prefix. It is set off in parentheses when naming the entire molecule.

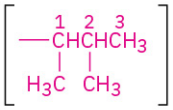


2,3-Dimethyl-6-(2-methylpropyl)decane

As a further example:

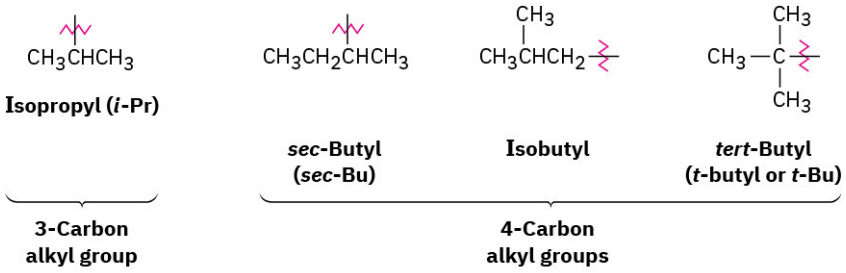


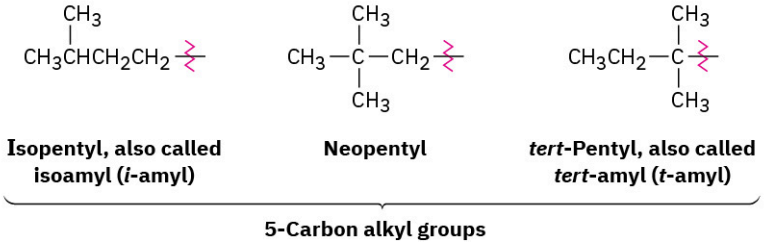
5-(1,2-Dimethylpropyl)-2-methylnonane



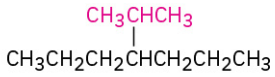
A 1,2-dimethylpropyl group

For historical reasons, some of the simpler branched-chain alkyl groups also have nonsystematic, common names, as noted earlier.





The common names of these simple alkyl groups are so well entrenched in the chemical literature that IUPAC rules make allowance for them. Thus, the following compound is properly named either 4-(1-methylethyl)heptane or 4-isopropylheptane. There's no choice but to memorize these common names; fortunately, there are only a few of them.



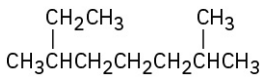
4-(1-Methylethyl)heptane or 4-Isopropylheptane

When writing an alkane name, the nonhyphenated prefix iso- is considered part of the alkyl-group name for alphabetizing purposes, but the hyphenated and italicized prefixes *sec*- and *tert*- are not. Thus, isopropyl and isobutyl are listed alphabetically under *i*, but *sec*-butyl and *tert*-butyl are listed under *b*.

WORKED EXAMPLE 3.2

Naming Alkanes

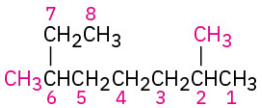
What is the IUPAC name for the following alkane?



Strategy

Find the longest continuous carbon chain in the molecule, and use that as the parent name. This molecule has a chain of eight carbons—octane—with two methyl substituents. (You have to turn corners to see it.) Numbering from the end nearer the first methyl substituent indicates that the methyls are at C2 and C6.

Solution



2,6-Dimethyloctane

WORKED EXAMPLE 3.3

Converting a Chemical Name into a Structure

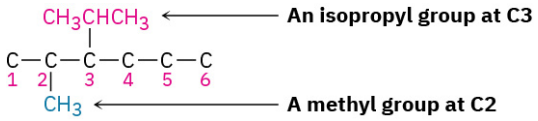
Draw the structure of 3-isopropyl-2-methylhexane.

Strategy

This is the reverse of **Worked Example 3.2** and uses a reverse strategy. Look at the parent name (hexane), and draw its carbon structure.

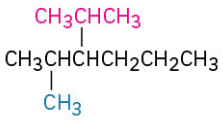
C–C–C–C–C–C **Hexane**

Next, find the substituents (3-isopropyl and 2-methyl), and place them on the proper carbons.



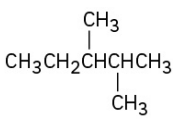
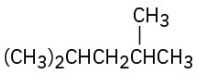
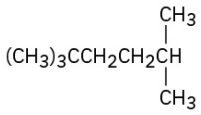
Finally, add hydrogens to complete the structure.

Solution



3-Isopropyl-2-methylhexane

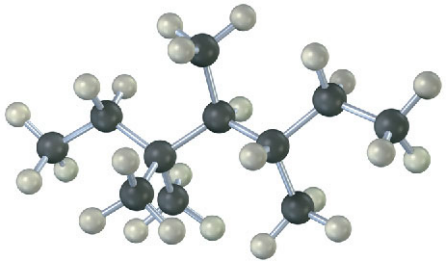
PROBLEM 3-11 Give IUPAC names for the following compounds:

(a) The three isomers of C_5H_{12} **(b)**  **(c)**  **(d)** 

PROBLEM 3-12 Draw structures corresponding to the following IUPAC names:
(a) 3,4-Dimethylnonane **(b)** 3-Ethyl-4,4-dimethylheptane **(c)** 2,2-Dimethyl-4-propyloctane **(d)** 2,2,4-Trimethylpentane

PROBLEM 3-13 Name the eight 5-carbon alkyl groups you drew in Problem 3-7.

PROBLEM 3-14 Give the IUPAC name for the following hydrocarbon, and convert the drawing into a skeletal structure.



3.5 Properties of Alkanes

Alkanes are sometimes referred to as *paraffins*, a word derived from the Latin *parum affinis*, meaning "little affinity." This term aptly describes their behavior, for alkanes show little chemical affinity for other substances and are chemically inert to most laboratory reagents. They are also relatively inert biologically and are not often involved in the chemistry of living organisms. Alkanes do, however, react with oxygen, halogens, and a few other substances under appropriate conditions.

Reaction with oxygen occurs during combustion in an engine or furnace when an alkane is used as a fuel. Carbon dioxide and water are formed as products, and a large amount of heat is released. For example, methane (natural gas) reacts with oxygen according to the equation

$$CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O + 890\ \text{kJ/mol (213 kcal/mol)}$$

The reaction of an alkane with Cl_2 occurs when a mixture of the two is irradiated with ultraviolet light (denoted $h\upsilon$, where υ is the Greek letter *nu*). Depending on the time allowed and the relative amounts of the two reactants, a sequential substitution of the alkane hydrogen atoms by chlorine occurs, leading to a mixture of chlorinated products. Methane, for instance, reacts with Cl_2 to yield a mixture of CH_3Cl, CH_2Cl_2, $CHCl_3$, and CCl_4. We'll look at this reaction in more detail in **Section 6.6**.

$$CH_4 + Cl_2 \xrightarrow{h\upsilon} CH_3Cl + HCl$$

$$CH_3Cl + Cl_2 \longrightarrow CH_2Cl_2 + HCl$$

$$CH_2Cl_2 + Cl_2 \longrightarrow CHCl_3 + HCl$$

$$CHCl_3 + Cl_2 \longrightarrow CCl_4 + HCl$$

Alkanes show regular increases in both boiling point and melting point as molecular weight increases (**FIGURE 3.5**), an effect due to the presence of weak dispersion forces between molecules (**Section 2.12**). Only when sufficient energy is applied to overcome these forces does the solid melt or liquid boil. As you might expect, dispersion forces increase as molecular size increases, accounting for the higher melting and boiling points of larger alkanes.

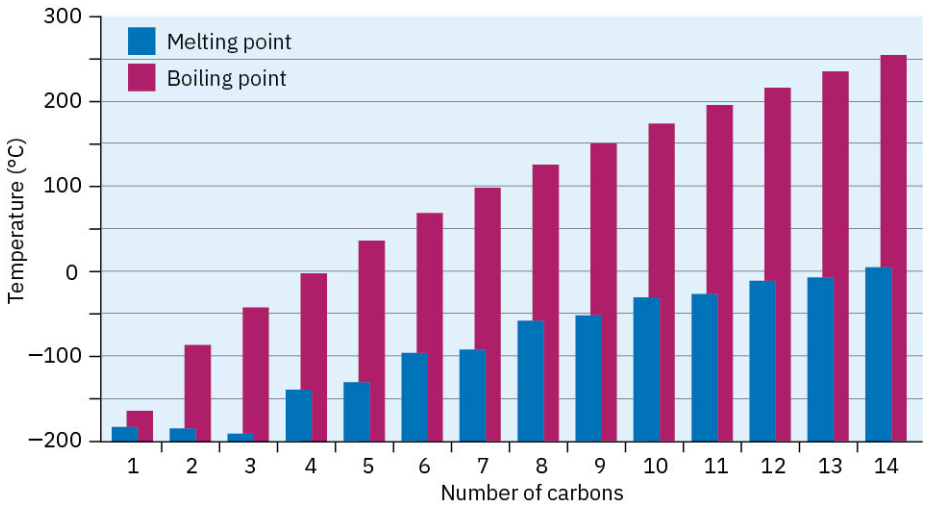


FIGURE 3.5 A plot of melting and boiling points versus number of carbon atoms for the C1–C14 straight-chain alkanes. There is a regular increase with molecular size.

Another effect seen in alkanes is that increased branching lowers an alkane's boiling point. Thus, pentane has no branches and boils at 36.1 °C, isopentane (2-methylbutane) has one branch and boils at 27.85 °C, and neopentane (2,2-dimethylpropane) has two branches and boils at 9.5 °C. Similarly, octane boils at 125.7 °C, whereas isooctane (2,2,4-trimethylpentane) boils at 99.3 °C. Branched-chain alkanes are lower-boiling because they are more nearly spherical than straight-chain alkanes, have smaller surface areas, and consequently have smaller dispersion forces.

3.6 Conformations of Ethane

Up until now, we've viewed molecules primarily in a two-dimensional way and have given little thought to any consequences that might arise from the spatial arrangement of atoms in molecules. Now it's time to add a third dimension to our study. **Stereochemistry** is the branch of chemistry concerned with the three-dimensional aspects of molecules. We'll see on many occasions in future chapters that the exact three-dimensional structure of a molecule is often crucial to determining its properties and biological behavior.

We know from **Section 1.5** that σ bonds are cylindrically symmetrical. In other words, the intersection of a plane cutting through a carbon–carbon single-bond orbital looks like a circle. Because of this cylindrical symmetry, rotation is possible around carbon–carbon bonds in open-chain molecules. In ethane, for instance, rotation around the C–C bond occurs freely, constantly changing the spatial relationships between the hydrogens on one carbon and those on the other (**FIGURE 3.6**).

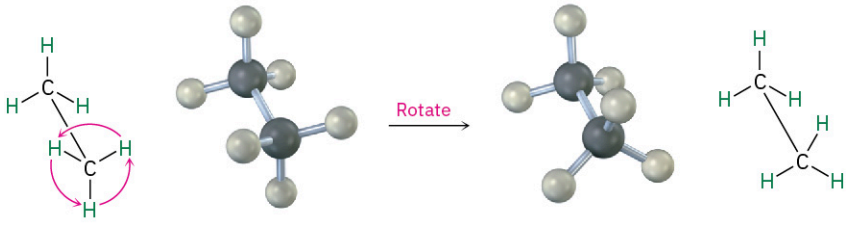


FIGURE 3.6 Rotation occurs around the carbon–carbon single bond in ethane because of σ bond cylindrical symmetry.

The different arrangements of atoms that result from bond rotation are called **conformations**, and molecules that have different arrangements are called **conformational isomers**, or **conformers**. Unlike constitutional isomers, however, different conformers often can't be isolated because they interconvert too rapidly.

Conformational isomers are represented in two ways, as shown in **FIGURE 3.7**. A **sawhorse representation** views the carbon–carbon bond from an oblique angle and indicates spatial orientation by showing all C–H bonds. A **Newman projection** views the carbon–carbon bond directly end-on and represents the two carbon atoms by a circle. Bonds attached to the front carbon are represented by lines to the center of the circle, and bonds attached to the rear carbon are represented by lines to the edge of the circle.

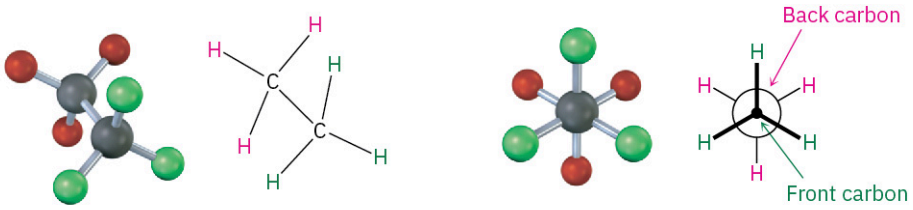
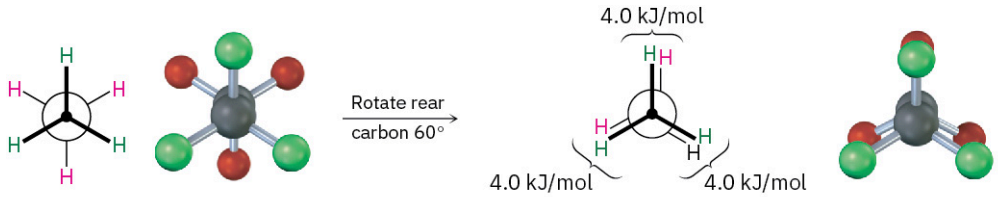


FIGURE 3.7 A sawhorse representation and a Newman projection of ethane. The sawhorse representation views the molecule from an oblique angle, while the Newman projection views the molecule end-on. Note that the molecular model of the Newman projection appears at first to have six atoms attached to a single carbon. Actually, the front carbon, with three attached **green atoms**, is directly in front of the rear carbon, with three attached **red atoms**.

Despite what we've just said, we actually don't observe perfectly free rotation in ethane. Experiments show that there is a small (12 kJ/mol; 2.9 kcal/mol) barrier to rotation and that some conformations are more stable than others. The lowest-energy, most stable conformation is the one in which all six C–H bonds are as far away from one another as possible—*staggered* when viewed end-on in a Newman projection. The highest-energy, least stable conformation is the one in which the six C–H bonds are as close as possible—*eclipsed* in a Newman projection. At any given instant, about 99% of ethane molecules have an approximately **staggered conformation** and only about 1% are near the **eclipsed conformation**.



The extra 12 kJ/mol of energy present in the eclipsed conformation of ethane is called **torsional strain**. Its cause has been the subject of controversy, but the major factor is an interaction between C–H bonding orbitals on one carbon and antibonding orbitals on the adjacent carbon, which stabilizes the staggered conformation relative to the eclipsed one. Because a total strain of 12 kJ/mol arises from three equal hydrogen–hydrogen eclipsing interactions, we can assign a value of approximately 4.0 kJ/mol (1.0 kcal/mol) to each single interaction. The barrier to rotation that results can be represented on a graph of potential energy versus degree of rotation, in which the angle between C–H bonds on the front and back carbons as viewed end-on (the *dihedral angle*) goes full circle from 0 to 360°. Energy minima occur at staggered conformations, and energy maxima occur at eclipsed conformations, as shown in FIGURE 3.8.

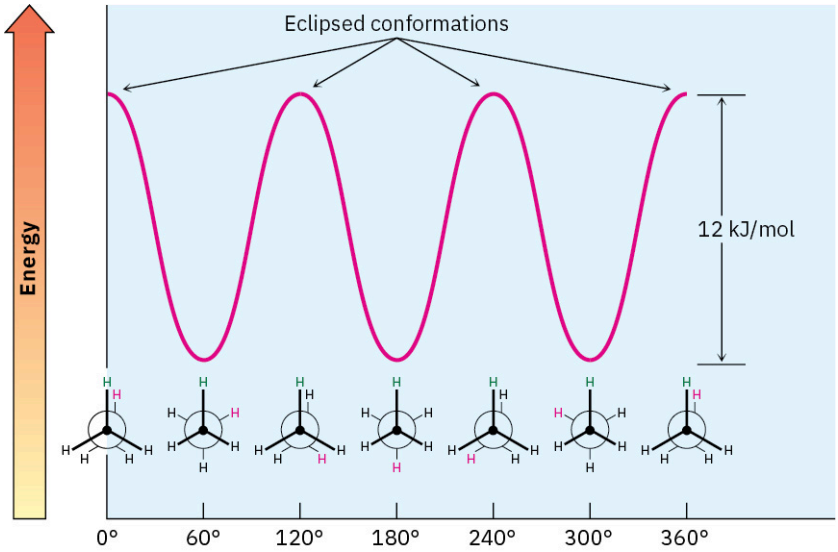


FIGURE 3.8 A graph of potential energy versus bond rotation in ethane. The staggered conformations are 12 kJ/mol lower in energy than the eclipsed conformations.

3.7 Conformations of Other Alkanes

Propane, the next-higher member in the alkane series, also has a torsional barrier that results in hindered rotation around the carbon–carbon bonds. The barrier is slightly higher in propane than in ethane—a total of 14 kJ/mol (3.4 kcal/mol) versus 12 kJ/mol.

The eclipsed conformation of propane has three interactions—two ethane-type hydrogen–hydrogen interactions and one additional hydrogen–methyl interaction. Since each eclipsing H ⟷ H interaction is the same as that in ethane and thus has an energy "cost" of 4.0 kJ/mol, we can assign a value of 14 – (2 × 4.0) = 6.0 kJ/mol (1.4 kcal/mol) to the eclipsing H ⟷ CH_3 interaction (FIGURE 3.9).

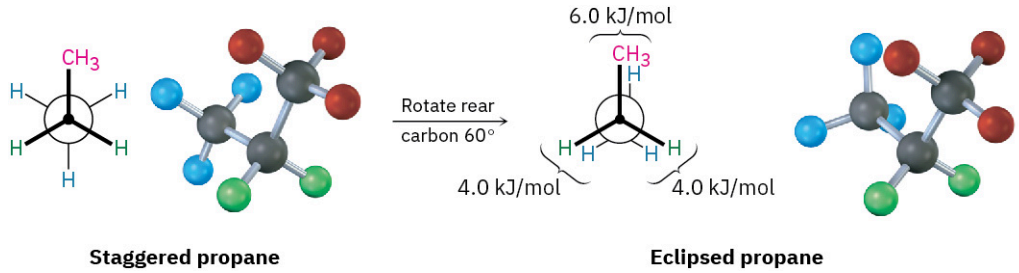
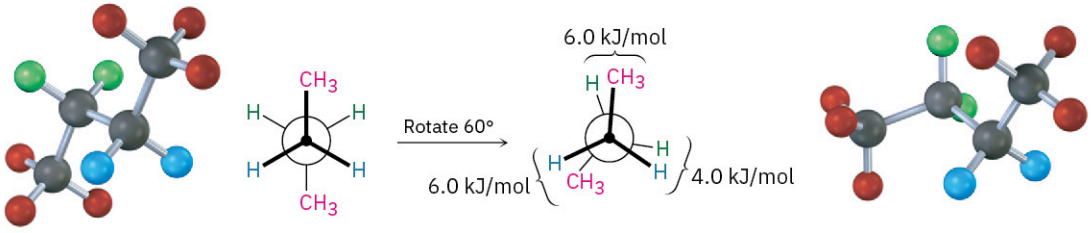


FIGURE 3.9 Newman projections of propane showing staggered and eclipsed conformations. The staggered conformer is lower in energy by 14 kJ/mol.

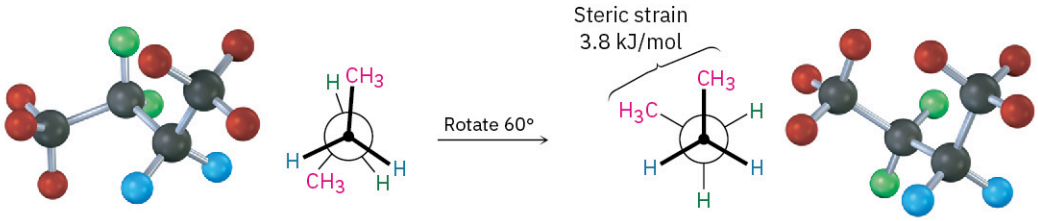
The conformational situation becomes more complex for larger alkanes because not all staggered conformations have the same energy and not all eclipsed conformations have the same energy. In butane, for instance, the lowest-energy arrangement, called the **anti conformation**, is the one in which the two methyl groups are as far apart as possible—180° away from each other. As rotation around the C2–C3 bond occurs, an eclipsed conformation is reached where there are two CH_3 ⟷ H interactions and one H ⟷ H interaction. Using the energy values derived previously from ethane and propane, this eclipsed conformation is more strained than the anti conformation by 2 × 6.0 kJ/mol + 4.0 kJ/mol (two CH_3 ⟷ H interactions plus one H ⟷ H interaction), for a total of 16 kJ/mol (3.8 kcal/mol).



Butane—anti conformation (0 kJ/mol)

Butane—eclipsed conformation (16 kJ/mol)

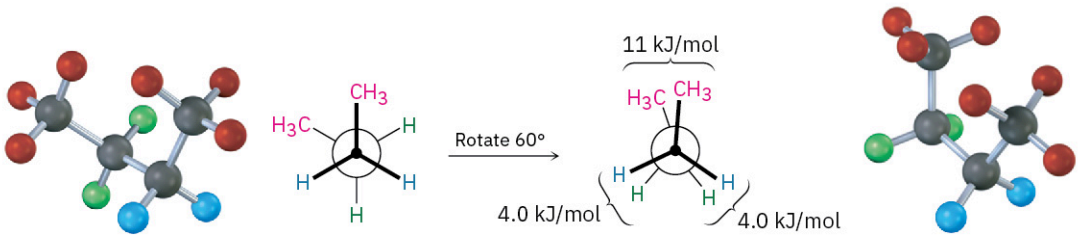
As bond rotation continues, an energy minimum is reached at the staggered conformation where the methyl groups are 60° apart. Called the **gauche conformation**, it lies 3.8 kJ/mol (0.9 kcal/mol) higher in energy than the anti conformation even though it has no eclipsing interactions. This energy difference occurs because the hydrogen atoms of the methyl groups are near one another in the gauche conformation, resulting in what is called *steric strain.* **Steric strain** is the repulsive interaction that occurs when atoms are forced closer together than their atomic radii allow. It's the result of trying to force two atoms to occupy the same space.



Butane—eclipsed conformation (16 kJ/mol)

Butane—gauche conformation (3.8 kJ/mol)

As the dihedral angle between the methyl groups approaches zero, an energy maximum is reached at a second eclipsed conformation. Because the methyl groups are forced even closer together than in the gauche conformation, both torsional strain and steric strain are present. A total strain energy of 19 kJ/mol (4.5 kcal/mol) has been estimated for this conformation, making it possible to calculate a value of 11 kJ/mol (2.6 kcal/mol) for the $CH_3 \longleftrightarrow CH_3$ eclipsing interaction: total strain of 19 kJ/mol minus the strain of two $H \longleftrightarrow H$ eclipsing interactions (2 × 4.0 kcal/mol) equals 11 kJ/mol.



Butane—gauche conformation (3.8 kJ/mol)

Butane—eclipsed conformation (19 kJ/mol)

After 0°, the rotation becomes a mirror image of what we've already seen: another gauche conformation is reached, another eclipsed conformation, and finally a return to the anti conformation. A plot of potential energy versus rotation about the C2–C3 bond is shown in FIGURE 3.10.

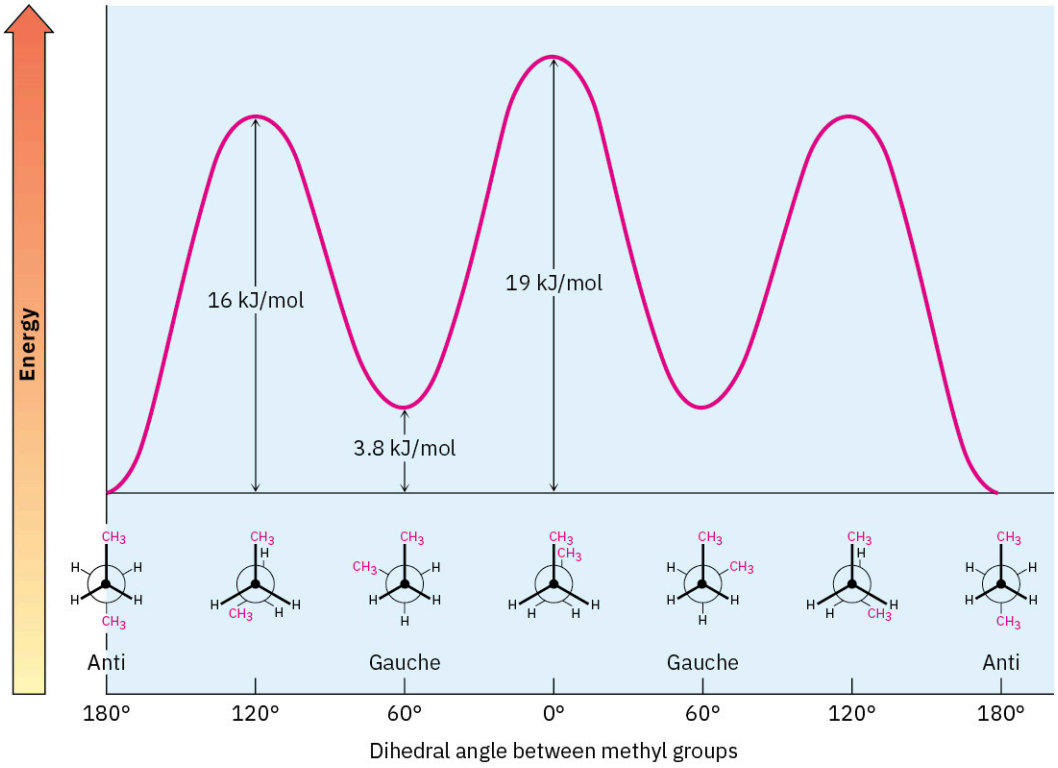


FIGURE 3.10 A plot of potential energy versus rotation for the C2–C3 bond in butane. The energy maximum occurs when the two methyl groups eclipse each other, and the energy minimum occurs when the two methyl groups are 180° apart (anti).

The notion of assigning definite energy values to specific interactions within a molecule is very useful, and we'll return to it in the next chapter. A summary of what we've seen thus far is given in TABLE 3.5.

The same principles just developed for butane apply to pentane, hexane, and all higher alkanes. The most favorable conformation for any alkane has the carbon–carbon bonds in staggered arrangements, with large substituents arranged anti to one another. A generalized alkane structure is shown in FIGURE 3.11.

TABLE 3.5 Energy Costs for Interactions in Alkane Conformers

Interaction	Cause	Energy cost	
		(kJ/mol)	(kcal/mol)
H ⟷ H eclipsed	Torsional strain	4.0	1.0
H ⟷ CH_3 eclipsed	Mostly torsional strain	6.0	1.4
CH_3 ⟷ CH_3 eclipsed	Torsional and steric strain	11.0	2.6
CH_3 ⟷ CH_3 gauche	Steric strain	3.8	0.9

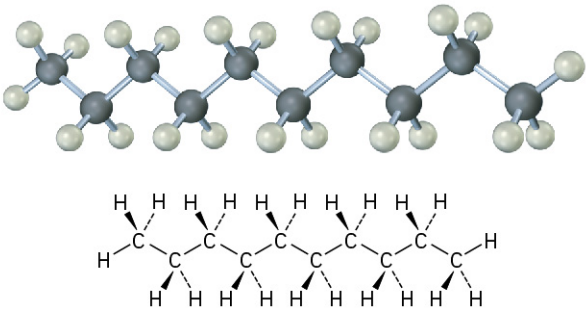


FIGURE 3.11 The most stable alkane conformation is the one in which all substituents are staggered and the carbon–carbon bonds are arranged anti, as shown in this model of decane.

One final point: saying that one particular conformer is "more stable" than another doesn't mean the molecule adopts and maintains only the more stable conformation. At room temperature, rotations around σ bonds occur so rapidly that all conformers are in equilibrium. At any given instant, however, a larger percentage of molecules will be found in a more stable conformation than in a less stable one.

WORKED EXAMPLE 3.4

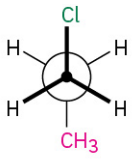
Newman Projections

Sight along the C1–C2 bond of 1-chloropropane, and draw Newman projections of the most stable and least stable conformations.

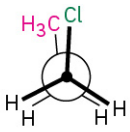
Strategy

The most stable conformation of a substituted alkane is generally a staggered one in which large groups have an anti relationship. The least stable conformation is generally an eclipsed one in which large groups are as close as possible.

Solution



Most stable (staggered)



Least stable (eclipsed)

PROBLEM 3-15 Make a graph of potential energy versus angle of bond rotation for propane, and assign values to the energy maxima.

PROBLEM 3-16 Sight along the C2–C1 bond of 2-methylpropane (isobutane).

(a) Draw a Newman projection of the most stable conformation.

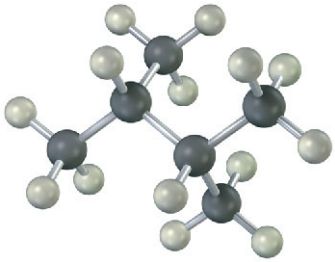
(b) Draw a Newman projection of the least stable conformation.

(c) Make a graph of energy versus angle of rotation around the C2–C1 bond.

(d) Assign relative values to the maxima and minima in your graph, given that an H ⟷ H eclipsing interaction costs 4.0 kJ/mol and an H ⟷ CH_3 eclipsing interaction costs 6.0 kJ/mol.

PROBLEM 3-17 Sight along the C2–C3 bond of 2,3-dimethylbutane, and draw a Newman projection of the most stable conformation.

PROBLEM 3-18 Draw a Newman projection along the C2–C3 bond of the following conformation of 2,3-dimethylbutane, and calculate a total strain energy:



CHEMISTRY MATTERS

Gasoline

British Foreign Minister Ernest Bevin once said that "The Kingdom of Heaven runs on righteousness, but the Kingdom of Earth runs on alkanes." (Actually, he said "runs on oil" not "runs on alkanes," but they're essentially the same.) By far, the major sources of alkanes are the world's natural gas and petroleum deposits. Laid down eons ago, these deposits are thought to be derived primarily from the decomposition of tiny single-celled marine organisms called foraminifera. Natural gas consists chiefly of methane but also contains ethane, propane, and butane. Petroleum is a complex mixture of hydrocarbons that must be separated into fractions and then further refined before it can be used.

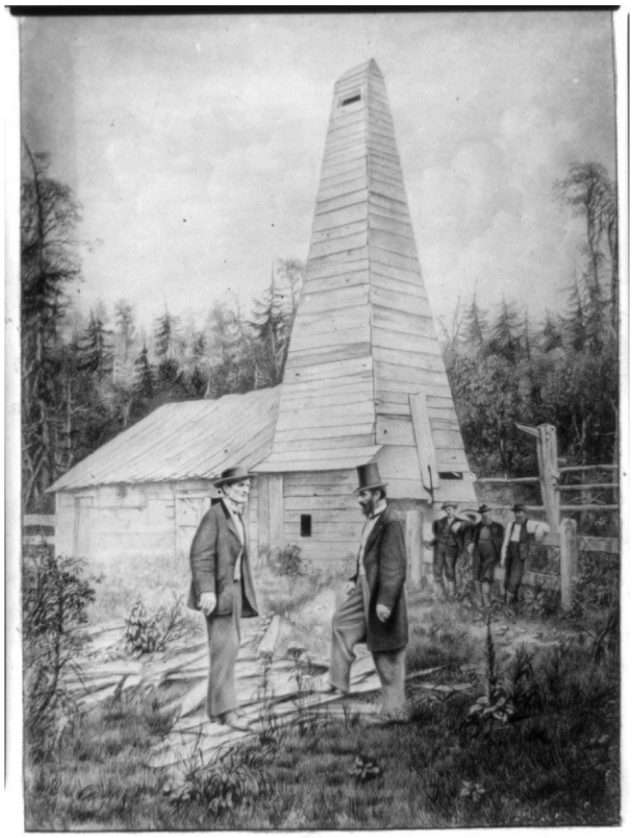
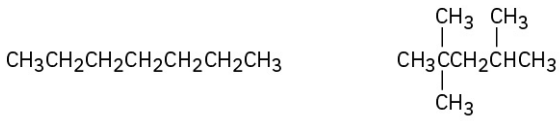


FIGURE 3.12 Gasoline is a finite resource. It won't be around forever. (credit: **"The first oil well" (https://www.loc.gov/item/2010649522/)** by Unknown/Library of Congress)

The petroleum era began in August 1859, when the world's first oil well was drilled by Edwin Drake near Titusville, Pennsylvania. The petroleum was distilled into fractions according to boiling point, but it was high-boiling kerosene, or lamp oil, rather than gasoline that was primarily sought. Literacy was becoming widespread at the time, and people wanted better light for reading than was available from candles. Gasoline was too volatile for use in lamps and was initially considered a waste by-product. The world has changed greatly since those early days, however, and it is now gasoline rather than lamp oil that is prized.

Petroleum refining begins by fractional distillation of crude oil into three principal cuts according to boiling point (bp): straight-run gasoline (bp 30–200 °C), kerosene (bp 175–300 °C), and heating oil, or diesel fuel (bp 275–400 °C). Further distillation under reduced pressure then yields lubricating oils and waxes and leaves a tarry residue of asphalt. The distillation of crude oil is only the first step in gasoline production, however. Straight-run gasoline turns out to be a poor fuel in automobiles because of engine knock, an uncontrolled combustion that can occur in a hot engine causing potentially serious damage.

The *octane number* of a fuel is the measure by which its antiknock properties are judged. It was recognized long ago that straight-chain hydrocarbons are far more prone to inducing engine knock than highly branched compounds. Heptane, a particularly bad fuel, is assigned a base value of 0 octane number, and 2,2,4-trimethylpentane, commonly known as isooctane, has a rating of 100.



Heptane (octane number = 0) **2,2,4-Trimethylpentane (octane number = 100)**

Because straight-run gasoline burns so poorly in engines, petroleum chemists have devised numerous methods for producing higher-quality fuels. One of these methods, *catalytic cracking*, involves taking the high-boiling kerosene cut (C_{11}–C_{14}) and "cracking" it into smaller branched molecules suitable for use in gasoline. Another process, called *reforming*, is used to convert C_6–C_8 alkanes to aromatic compounds such as benzene and toluene, which have substantially higher octane numbers than alkanes. The final product that goes in your tank has an approximate composition of 15% C_4–C_8 straight-chain alkanes, 25% to 40% C_4–C_{10} branched-chain alkanes, 10% cyclic alkanes, 10% straight-chain and cyclic alkenes, and 25% arenes (aromatics).

Key Terms

- **alcohol**
- **aldehyde**
- **aliphatic**
- **alkane**
- alkene
- **alkyl group**
- **alkyl halide**
- alkyne
- **amide**
- **amine**
- **anti conformation**
- arene
- **branched-chain alkane**
- **carbonyl group**
- **carboxylic acid**
- **conformation**
- **conformational isomer**
- **conformer**
- **constitutional isomer**
- **eclipsed conformation**
- **ester**
- **ether**
- **functional group**
- **gauche conformation**
- **hydrocarbon**
- **isomer**
- **ketone**
- **Newman projection**
- nitrile
- **R group**
- **saturated**
- **sawhorse representation**
- **staggered conformation**
- **stereochemistry**
- **steric strain**
- **straight-chain alkane**
- **substituent**
- **sulfide**
- **thiol**
- **torsional strain**

Summary

Alkanes are relatively unreactive and rarely involved in chemical reactions, but they nevertheless provide a useful vehicle for introducing some important general ideas. In this chapter, we've used alkanes to introduce the basic approach to naming organic compounds and to take an initial look at some of the three-dimensional aspects of molecules.

A **functional group** is a group of atoms within a larger molecule that has a characteristic chemical reactivity. Because functional groups behave in approximately the same way in all molecules where they occur, the chemical reactions of an organic molecule are largely determined by its functional groups.

Alkanes are a class of **saturated hydrocarbons** with the general formula C_nH_{2n+2}. They contain no functional groups, are relatively inert, and can be either **straight-chain** (*normal*) or **branched**. Alkanes are named by a series of IUPAC rules of nomenclature. Compounds that have the same chemical formula but different

structures are called **isomers**. More specifically, compounds such as butane and isobutane, which differ in their connections between atoms, are called **constitutional isomers**.

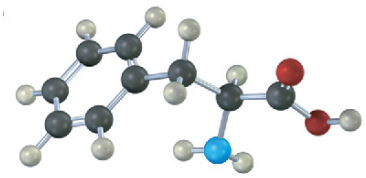
Carbon–carbon single bonds in alkanes are formed by σ overlap of carbon sp^3 hybrid orbitals. Rotation is possible around σ bonds because of their cylindrical symmetry, and alkanes therefore exist in a large number of rapidly interconverting **conformations**. **Newman projections** make it possible to visualize the spatial consequences of bond rotation by sighting directly along a carbon–carbon bond axis. Not all alkane conformations are equally stable. The **staggered conformation** of ethane is 12 kJ/mol (2.9 kcal/mol) more stable than the **eclipsed conformation** because of **torsional strain**. In general, any alkane is most stable when all its bonds are staggered.

Additional Problems

Visualizing Chemistry

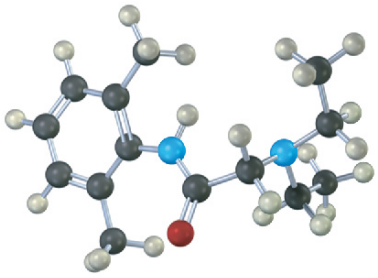
PROBLEM 3-19 Identify the functional groups in the following substances, and convert each drawing into a molecular formula (red = O, blue = N).

(a)



Phenylalanine

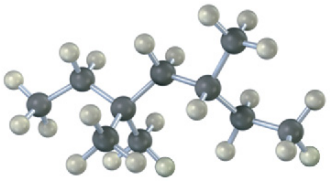
(b)



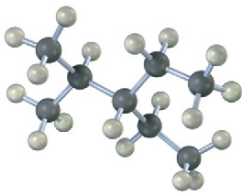
Lidocaine

PROBLEM 3-20 Give IUPAC names for the following alkanes, and convert each drawing into a skeletal structure.

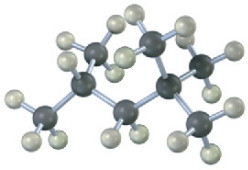
(a)



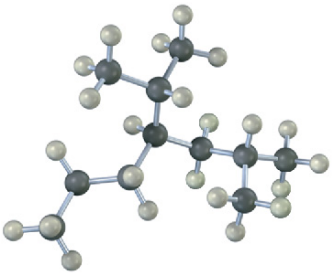
(b)



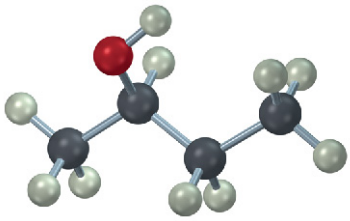
(c)



(d)

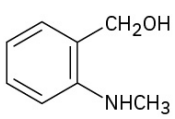


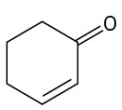
PROBLEM 3-21 Draw a Newman projection along the C2–C3 bond of the following conformation of 2-butanol.

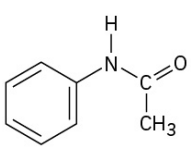


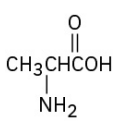
Functional Groups

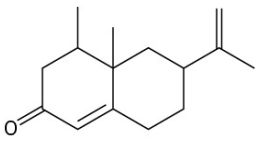
PROBLEM 3-22 Locate and identify the functional groups in the following molecules.

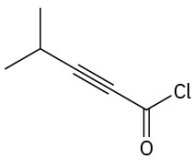
(a) 

(b) 

(c) 

(d) 

(e) 

(f) 

PROBLEM 3-23 Propose structures that meet the following descriptions:

(a) A ketone with five carbons **(b)** A four-carbon amide **(c)** A five-carbon ester
(d) An aromatic aldehyde **(e)** A keto ester **(f)** An amino alcohol

PROBLEM 3-24 Propose structures for the following:

(a) A ketone, C_4H_8O **(b)** A nitrile, C_5H_9N **(c)** A dialdehyde, $C_4H_6O_2$
(d) A bromoalkene, $C_6H_{11}Br$ **(e)** An alkane, C_6H_{14} **(f)** *cyclic* saturated hydrocarbon, C_6H_{12}
(g) A diene (dialkene), C_5H_8 **(h)** A keto alkene, C_5H_8O

PROBLEM 3-25 Predict the hybridization of the carbon atom in each of the following functional groups:

(a) Ketone **(b)** Nitrile **(c)** Carboxylic acid

PROBLEM 3-26 Draw the structures of the following molecules:

(a) Biacetyl, $C_4H_6O_2$, a substance with the aroma of butter; it contains no rings or carbon–carbon multiple bonds.
(b) Ethylenimine, C_2H_5N, a substance used in the synthesis of melamine polymers; it contains no multiple bonds.
(c) Glycerol, $C_3H_8O_3$, a substance isolated from fat and used in cosmetics; it has an –OH group on each carbon.

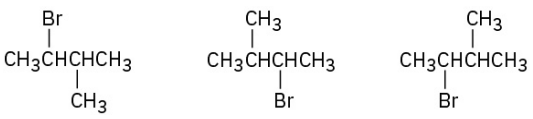
Isomers

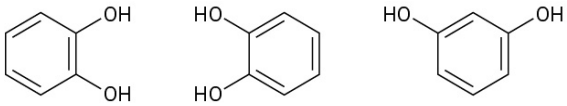
PROBLEM 3-27 Draw structures that meet the following descriptions (there are many possibilities):

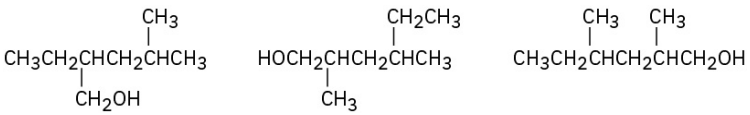
(a) Three isomers with the formula C_8H_{18} **(b)** Two isomers with the formula $C_4H_8O_2$

PROBLEM 3-28 Draw structures of the nine isomers of C_7H_{16}.

PROBLEM 3-29 In each of the following sets, which structures represent the same compound and which represent different compounds?

(a) 

(b) 

(c) 

PROBLEM 3-30 Seven constitutional isomers have the formula $C_4H_{10}O$. Draw as many as you can.

PROBLEM Draw as many compounds as you can that fit the following descriptions:

3-31 **(a)** Alcohols with formula $C_4H_{10}O$ **(b)** Amines with formula $C_5H_{13}N$
(c) Ketones with formula $C_5H_{10}O$ **(d)** Aldehydes with formula $C_5H_{10}O$
(e) Esters with formula $C_4H_8O_2$ **(f)** Ethers with formula $C_4H_{10}O$

PROBLEM Draw compounds that contain the following:

3-32 **(a)** A primary alcohol **(b)** A tertiary nitrile **(c)** A secondary thiol
(d) Both primary and secondary alcohols **(e)** An isopropyl group **(f)** A quaternary carbon

Naming Compounds

PROBLEM Draw and name all monobromo derivatives of pentane, $C_5H_{11}Br$.

3-33

PROBLEM Draw and name all monochloro derivatives of 2,5-dimethylhexane, $C_8H_{17}Cl$.

3-34

PROBLEM Draw structures for the following:

3-35 **(a)** 2-Methylheptane **(b)** 4-Ethyl-2,2-dimethylhexane **(c)** 4-Ethyl-3,4-dimethyloctane
(d) 2,4,4-Trimethylheptane **(e)** 3,3-Diethyl-2,5-dimethylnonane
(f) 4-Isopropyl-3-methylheptane

PROBLEM Draw a compound that:

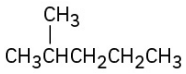
3-36 **(a)** Has only primary and tertiary carbons **(b)** Has no secondary or tertiary carbons
(c) Has no secondary or tertiary carbons

PROBLEM Draw a compound that:

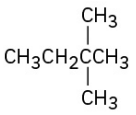
3-37 **(a)** Has nine primary hydrogens **(b)** Has only primary hydrogens

PROBLEM Give IUPAC names for the following compounds:

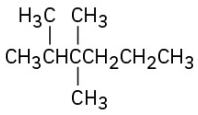
3-38 **(a)**



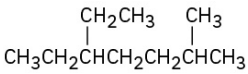
(b)



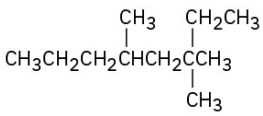
(c)



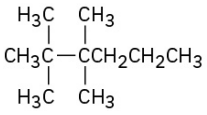
(d)



(e)



(f)



PROBLEM Name the five isomers of C_6H_{14}.

3-39

PROBLEM Explain why each of the following names is incorrect:

3-40 **(a)** 2,2-Dimethyl-6-ethylheptane **(b)** 4-Ethyl-5,5-dimethylpentane
(c) 3-Ethyl-4,4-dimethylhexane **(d)** 5,5,6-Trimethyloctane **(e)** 2-Isopropyl-4-methylheptane

PROBLEM Propose structures and give IUPAC names for the following:

3-41 **(a)** A diethyldimethylhexane **(b)** A (3-methylbutyl)-substituted alkane

Conformations

PROBLEM Consider 2-methylbutane (isopentane). Sighting along the C2–C3 bond:

3-42 **(a)** Draw a Newman projection of the most stable conformation.
(b) Draw a Newman projection of the least stable conformation.
(c) If a $CH_3 \longleftrightarrow CH_3$ eclipsing interaction costs 11 kJ/mol (2.5 kcal/mol) and a $CH_3 \longleftrightarrow CH_3$ gauche interaction costs 3.8 kJ/mol (0.9 kcal/mol), make a quantitative plot of energy versus rotation about the C2–C3 bond.

PROBLEM What are the relative energies of the three possible staggered conformations around the C2–C3 bond in 2,3-dimethylbutane? (See Problem 3-42.)

3-43

PROBLEM Construct a qualitative potential-energy diagram for rotation about the C–C bond of

3-44 1,2-dibromoethane. Which conformation would you expect to be most stable? Label the anti and gauche conformations of 1,2-dibromoethane.

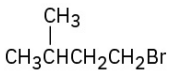
PROBLEM 3-45 Which conformation of 1,2-dibromoethane (Problem 3-44) would you expect to have the largest dipole moment? The observed dipole moment of 1,2-dibromoethane is $\mu = 1.0$ D. What does this tell you about the actual conformation of the molecule?

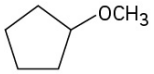
PROBLEM 3-46 Draw the most stable conformation of pentane, using wedges and dashes to represent bonds coming out of the paper and going behind the paper, respectively.

PROBLEM 3-47 Draw the most stable conformation of 1,4-dichlorobutane, using wedges and dashes to represent bonds coming out of the paper and going behind the paper, respectively.

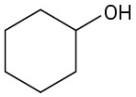
General Problems

PROBLEM 3-48 For each of the following compounds, draw an isomer that has the same functional groups.

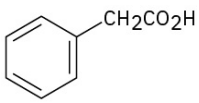
(a) 

(b) 

(c) $CH_3CH_2CH_2C{\equiv}N$

(d) 

(e) CH_3CH_2CHO

(f) 

PROBLEM 3-49 Malic acid, $C_4H_6O_5$, has been isolated from apples. Because this compound reacts with 2 molar equivalents of base, it is a dicarboxylic acid.

(a) Draw at least five possible structures.

(b) If malic acid is a secondary alcohol, what is its structure?

PROBLEM 3-50 Formaldehyde, $H_2C{=}O$, is known to all biologists because of its usefulness as a tissue preservative. When pure, formaldehyde trimerizes to give trioxane, $C_3H_6O_3$, which, surprisingly enough, has no carbonyl groups. Only one monobromo derivative ($C_3H_5BrO_3$) of trioxane is possible. Propose a structure for trioxane.

PROBLEM 3-51 The barrier to rotation about the C–C bond in bromoethane is 15 kJ/mol (3.6 kcal/mol).

(a) What energy value can you assign to an H ↔ Br eclipsing interaction?

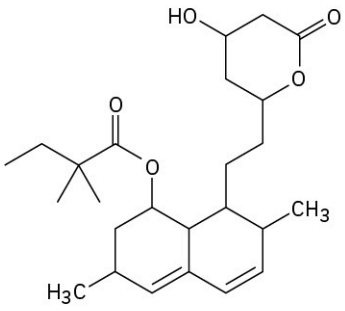
(b) Construct a quantitative diagram of potential energy versus bond rotation for bromoethane.

PROBLEM 3-52 Increased substitution around a bond leads to increased strain. Take the four substituted butanes listed below, for example. For each compound, sight along the C2–C3 bond and draw Newman projections of the most stable and least stable conformations. Use the data in Table 3.5 to assign strain-energy values to each conformation. Which of the eight conformations is most strained? Which is least strained?

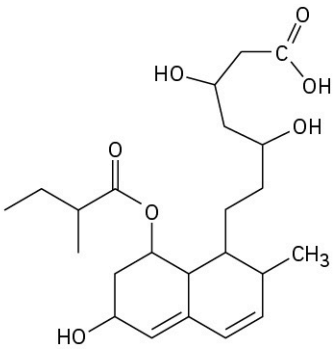
(a) 2-Methylbutane **(b)** 2,2-Dimethylbutane **(c)** 2,3-Dimethylbutane

(d) 2,2,3-Trimethylbutane

PROBLEM 3-53 The cholesterol-lowering agents called *statins*, such as simvastatin (Zocor) and pravastatin (Pravachol), are among the most widely prescribed drugs in the world, with annual sales estimated at approximately $25 billion. Identify the functional groups in both, and tell how the two substances differ.

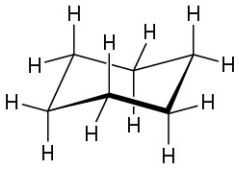


Simvastatin (Zocor)

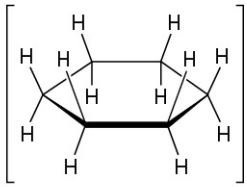


Pravastatin (Pravachol)

PROBLEM 3-54 In the next chapter we'll look at *cycloalkanes*—saturated cyclic hydrocarbons—and we'll see that the molecules generally adopt puckered, nonplanar conformations. Cyclohexane, for instance, has a puckered shape like a lounge chair rather than a flat shape. Why?

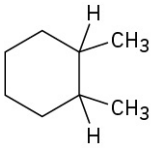


Nonplanar cyclohexane



Planar cyclohexane

PROBLEM 3-55 We'll see in the next chapter that there are two isomeric substances, both named 1,2-dimethylcyclohexane. Explain.



1,2-Dimethylcyclohexane